

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO5 ASSESSMENT TOOL 1 – Research Paper Review / Literature Survey

Course Code:	CSA10	Course Title:	Software Engineering
CO Assessed:	CO5	Assessment:	Assessment Tool 1 – Research Paper Review / Literature Survey
Weightage:	20%	Total Marks:	25
CO5 Statement:	Evaluate emerging technologies and societal impacts, maintaining and evolving software systems responsibly		
Student Name:	S Jefna Princy	Reg. No.:	192572058
Date:	24-08-2026	Duration:	60 minutes

Code of Conduct

I S Jefna Princy- 192572058 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Research Paper Review / Literature Survey

Instructions to Students:

Each student is assigned an individual problem statement. Based on the assigned problem statement, **conduct a literature survey and review relevant research papers** to understand the existing approaches, techniques, technologies, limitations, and research gaps related to the problem. The literature survey should include:

1. Identify and select **relevant research papers** related to the assigned problem statement from reputed journals, conferences, or scholarly databases.
2. Summarize the **objectives, methodologies, technologies, datasets, and major findings** of the selected research papers.
3. Compare the existing approaches based on suitable parameters such as **accuracy, performance, scalability, efficiency, methodology, or other problem-specific metrics**.
4. Identify the **limitations and challenges** of the existing research approaches.
5. Analyze the literature to identify the **research gap** or unresolved issues related to the assigned problem.
6. Discuss the **possible improvements or future research directions** based on the identified research gap.
7. Prepare a **comparative literature survey table** containing details such as author/year, research problem, methodology, dataset/tools, key findings, limitations, and research gap.
8. Provide proper **citations and references** using a consistent referencing style.

Deliverable: Submit a **Research Paper Review / Literature Survey Report** containing the selected research papers, individual paper reviews, comparative analysis, identified research gaps, future directions, and properly formatted references.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Selection & Review of Research Papers(25%)	Selects highly relevant, recent, credible papers and provides comprehensive reviews of objectives, methodology, tools, datasets, and findings	Selects relevant papers and provides good reviews with minor gaps	Selects moderately relevant papers with basic reviews	Papers are irrelevant or reviews are incomplete
Comparative Analysis(20%)	Provides a detailed comparison of methodologies, tools, datasets, results, and performance using a clear comparison table	Provides a good comparison with relevant parameters	Provides limited comparison with few parameters	Little or no meaningful comparison
Critical Analysis & Research Gap(25%)	Clearly analyzes limitations and identifies significant, well-supported research gaps	Identifies relevant limitations and research gaps	Identifies basic gaps with limited analysis	Research gaps are unclear or missing
Future Directions & Relevance(15%)	Proposes practical and innovative future directions directly relevant to the assigned problem statement	Proposes relevant future directions	Provides general future suggestions	Future directions are missing or unrelated
Documentation , Citations & References(15%)	Excellent academic structure, clear presentation, accurate citations, and consistent referencing	Well-organized with minor formatting/citation issues	Adequately documented with some errors	Poorly organized with inadequate or incorrect references

Criteria	Mark
Selection & Review of Research Papers(25%)	/5
Comparative Analysis(20%)	/5
Critical Analysis & Research Gap(25%)	/5

Future Directions & Relevance(15%)	/5
Documentation, Citations & References(15%)	/5
TOTAL	/25

RESEARCH PAPER REVIEW / LITERATURE SURVEY

Title

Literature Review of AI-Driven Smart Urban Noise Pollution Monitoring and Mitigation Systems

1. Introduction

Urban noise pollution is an increasing environmental problem caused by transportation, construction, industries, and other human activities. Traditional noise monitoring systems generally depend on fixed monitoring stations or periodic manual measurements. These approaches may not provide sufficient spatial and temporal coverage for rapidly changing urban environments.

Recent research has explored the use of **Internet of Things (IoT), Artificial Intelligence (AI), Machine Learning (ML), cloud computing, edge computing, and Geographic Information Systems (GIS)** for continuous and intelligent noise monitoring.

IoT sensors can collect noise measurements continuously from different locations, while cloud platforms can store and process large volumes of environmental data. Machine learning techniques can then identify patterns, classify noise sources, predict future noise levels, and detect abnormal measurements. GIS can further represent noise levels spatially and help authorities identify noise pollution hotspots.

Therefore, this literature review analyzes existing research approaches and identifies the research gap for developing an integrated **AI-driven smart urban noise pollution monitoring and mitigation system**.

2. Objectives of the Literature Survey

The main objectives are:

1. To study existing IoT-based urban noise monitoring systems.

2. To analyze AI and machine learning techniques used for noise prediction and classification.
3. To examine the use of GIS for identifying noise pollution hotspots.
4. To compare different sensing, processing, and prediction approaches.
5. To identify limitations of existing research.
6. To identify research gaps in intelligent urban noise monitoring.
7. To suggest possible future improvements for an integrated AI-IoT-GIS system.

3. Review of Selected Research Papers

We can use **8 papers**, which is enough to make the literature review look strong without making it unnecessarily huge.

Paper 1 – RaveGuard

Authors: Balestrieri et al.

Year: 2020

Title: *RaveGuard: A Noise Monitoring Platform Using Low-End Microphones and Machine Learning*

The researchers developed RaveGuard, a low-cost noise monitoring platform using inexpensive microphones and machine learning. The system was tested in Bologna, Italy, alongside a professional sound-level meter. Machine learning was used to improve the accuracy of low-cost sensors. The study reported a relative error of approximately **2.24%** compared with professional instruments.

Contribution: Demonstrates that low-cost IoT sensing combined with machine learning can support large-scale urban noise monitoring.

Limitation: The system mainly focuses on improving measurement accuracy rather than providing a complete AI-GIS decision-support platform.

Paper 2 – Deep Learning for Urban Noise Monitoring

Authors: Renaud, Karam, Salomon & Couturier

Year: 2023

Title: *Deep Learning and Gradient Boosting for Urban Environmental Noise Monitoring in Smart Cities*

This study investigated machine learning and deep learning models for forecasting urban environmental noise. Models including **LSTM, CNN-LSTM, Transformer, Temporal Fusion Transformer and LightGBM** were studied using real noise data collected from parking meters in an English city. The researchers achieved short-term and long-term noise forecasting and also investigated anomaly detection.

Contribution: Shows that deep learning can predict future noise levels and detect abnormal noise data.

Limitation: The research focuses mainly on prediction and anomaly detection rather than integrating prediction with GIS-based hotspot visualization and mitigation.

Paper 3 – Environmental Noise Classification

Authors: Albaji & Rashid

Year: 2023

Title: *Investigation on Machine Learning Approaches for Environmental Noise Classifications*

This research investigated different machine learning approaches for classifying environmental sounds. The objective was to identify suitable ML algorithms for recognizing different types of environmental noise in smart-city environments.

Contribution: Demonstrates the usefulness of machine learning for identifying and classifying environmental noise.

Limitation: Classification alone does not provide continuous city-wide monitoring, spatial hotspot identification, or complete mitigation support.

Paper 4 – TinyML Noise Anomaly Detection

Authors: Hammad, Iskandaryan & Trilles

Year: 2023

Title: *An Unsupervised TinyML Approach Applied to the Detection of Urban Noise Anomalies Under the Smart Cities Environment*

The researchers developed a TinyML approach using an **LSTM Autoencoder** for detecting abnormal noise measurements in urban IoT sensor networks. The trained model was deployed on an ESP32 microcontroller. The study reported very high anomaly-detection accuracy and demonstrated that AI processing can be performed directly on resource-constrained devices.

Contribution: Shows the potential of edge AI/TinyML for fast anomaly detection with low computational requirements.

Limitation: The approach concentrates on anomaly detection and does not provide a complete cloud-GIS-dashboard-based management system.

Paper 5 – Smart City Noise Classification

Authors: Baucas & Spachos

Year: 2024

Title: *Edge-Based Data Sensing and Processing Platform for Urban Noise Classification*

The researchers proposed a low-cost IoT-based platform for urban noise classification. The system used sensing hardware and edge processing before sending results to a cloud server. The study evaluated classification accuracy, power consumption, and latency between edge and cloud processing.

Contribution: Demonstrates the benefits of combining IoT sensing, edge computing and cloud processing.

Limitation: The work primarily addresses noise classification rather than complete prediction, GIS hotspot analysis, alert generation and urban planning.

Paper 6 – Road Traffic Noise Monitoring

Year: 2023

Title: *Road Traffic Noise Monitoring in a Smart City: Sensor and Model-Based Approach*

This study developed a road traffic noise model using traffic volume and vehicle-speed information. The model was evaluated using traffic radar and noise-sensor measurements in Aveiro, Portugal. The reported mean absolute percentage errors were no more than **5.8%**.

Contribution: Demonstrates that traffic information can be combined with acoustic modelling to estimate road traffic noise.

Limitation: The approach is specifically focused on traffic noise and does not cover other important urban sources such as construction and industrial activities.

Paper 7 – GeoAI for Noise Pollution

Year: 2024

Title: *Exploring Multi-Pollution Variability in the Urban Environment: Geospatial AI-Driven Modeling of Air and Noise*

This study used **Geospatial AI (GeoAI)** to model air and noise pollution in Tehran. The researchers used spatial data and machine learning techniques including CatBoost and optimization algorithms to produce pollution susceptibility maps. The optimized models reduced noise-pollution prediction error compared with the standalone CatBoost model.

Contribution: Demonstrates the importance of combining machine learning with geospatial analysis for identifying pollution patterns.

Limitation: The study focuses on spatial modelling rather than a real-time IoT monitoring and alert system.

Paper 8 – Chennai Noise Monitoring

Year: 2024

Title: *Evaluating Variations in Environmental Noise Pollution of Chennai City Using a Mobile Monitoring Technique*

This research investigated environmental noise levels across Chennai using a mobile monitoring approach. Measurements were collected while moving through the city, allowing researchers to examine spatial and temporal variations in noise exposure. The study reported noise exposure levels between approximately **74 and 85 dB(A)** for urban travellers and highlighted the importance of extensive monitoring for urban planning.

Contribution: Provides location-specific evidence of urban noise variation and demonstrates the need for extensive monitoring in Chennai.

Limitation: Mobile monitoring is useful for surveys but does not provide continuous automated IoT monitoring and AI-based prediction.

4. Comparative Literature Survey Table

Author/Year	Main Approach	Technology/Method	Major Finding	Limitation
Balestrieri et al., 2020	Low-cost noise monitoring	IoT + ML	Low-cost sensors can achieve good accuracy	Limited integrated decision support
Renaud et al., 2023	Noise prediction	LSTM, CNN-LSTM, Transformer, LightGBM	Accurate short/long-term prediction possible	Limited GIS/mitigation integration
Albaji & Rashid, 2023	Noise classification	Machine Learning	ML can classify environmental sounds	Does not provide complete monitoring platform
Hammad et al., 2023	Noise anomaly detection	TinyML + LSTM Autoencoder	Efficient edge anomaly detection	Mainly focuses on anomalies
Baucas & Spachos, 2024	Urban sound classification	IoT + Edge + Cloud	Edge processing can reduce latency	Limited prediction and GIS functionality
2023 Traffic Noise Study	Traffic noise estimation	Traffic sensors + acoustic model	Traffic data can estimate noise levels	Mainly traffic-focused
2024 GeoAI Study	Pollution hotspot modelling	GIS + CatBoost + optimization	GeoAI can identify spatial pollution patterns	Not a complete real-time IoT system
Chennai Study, 2024	Urban noise measurement	Mobile monitoring	Chennai has significant	Monitoring is not

			spatial noise variation	continuously automated
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5. Comparative Analysis

Existing research shows that different technologies solve different parts of the urban noise problem.

IoT-based approaches

IoT systems are useful for continuous collection of noise measurements from multiple locations. However, sensor accuracy, maintenance, connectivity and deployment cost remain important challenges.

Machine Learning approaches

ML and deep learning models such as LSTM, CNN-LSTM, Transformer, LightGBM and autoencoders can predict noise levels, classify sound sources and identify anomalies. However, model performance depends strongly on the availability and quality of training data.

Edge Computing

Edge-based systems can process data close to the sensor. This can reduce communication delay and enable faster detection of abnormal noise levels. However, edge devices have limited computational resources.

GIS approaches

GIS provides an effective way of visualizing spatial noise patterns and identifying high-noise areas. However, many GIS-based studies use collected datasets rather than continuous real-time IoT data.

Cloud Computing

Cloud platforms provide scalable storage and processing for large sensor networks. However, cloud-only processing can introduce communication latency, making a hybrid edge-cloud architecture more suitable for real-time alerts.

6. Critical Analysis and Research Gap

The reviewed research demonstrates significant progress in IoT-based monitoring, AI prediction, edge processing and GIS-based environmental analysis. However, these technologies are frequently implemented as **separate solutions**.

The major research gaps identified are:

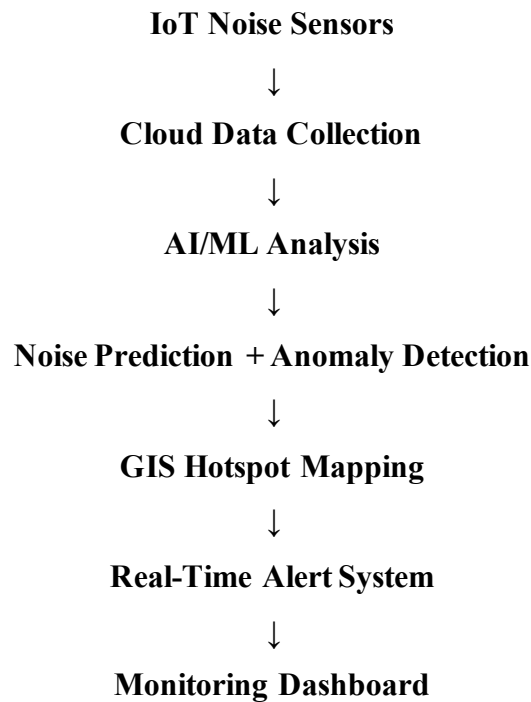
1. **Lack of complete integration:** Many studies focus on only one component such as sensing, prediction, classification or GIS.
2. **Limited real-time monitoring:** Some approaches depend on periodic, mobile or previously collected datasets.
3. **Limited multi-source analysis:** Many systems focus mainly on traffic noise instead of combining traffic, construction, industrial and other sources.
4. **Limited real-time alert mechanisms:** Prediction models do not always connect directly to automated environmental alerts.
5. **Limited GIS-AI integration:** GIS is often used for visualization, while AI prediction operates separately.
6. **Sensor reliability issues:** Low-cost sensors may produce inaccurate or abnormal readings and require calibration and anomaly detection.
7. **Limited mitigation support:** Existing studies frequently identify or predict noise but provide limited support for selecting appropriate mitigation actions.

Identified Research Gap

Therefore, there is a need for an **integrated AI-driven IoT-GIS platform** that can continuously collect noise data, process it using cloud/edge computing, predict future noise levels, identify hotspots using GIS, detect abnormal readings and automatically generate alerts for authorities.

7. Proposed Research Direction

Based on the identified research gap, the proposed system can combine:



The proposed architecture would provide both **real-time monitoring** and **predictive analysis** rather than simply recording historical noise measurements.

8. Future Research Directions

Future research can focus on:

1. Developing larger and geographically diverse noise datasets.
2. Combining multiple sensor types and environmental parameters.
3. Improving calibration methods for low-cost sensors.
4. Using lightweight AI models for edge devices.
5. Integrating real-time GIS hotspot mapping.
6. Developing automatic alerts when noise limits are exceeded.
7. Combining noise prediction with traffic and weather information.
8. Developing AI-based recommendations for noise mitigation.
9. Providing mobile and web dashboards for authorities.
10. Evaluating the system using real-world urban deployments.

9. Conclusion

The literature review demonstrates that **IoT, AI, machine learning, edge computing, cloud computing and GIS** can significantly improve urban noise monitoring. Existing studies have successfully addressed low-cost sensing, noise classification, prediction, anomaly detection, traffic-noise modelling and spatial pollution analysis.